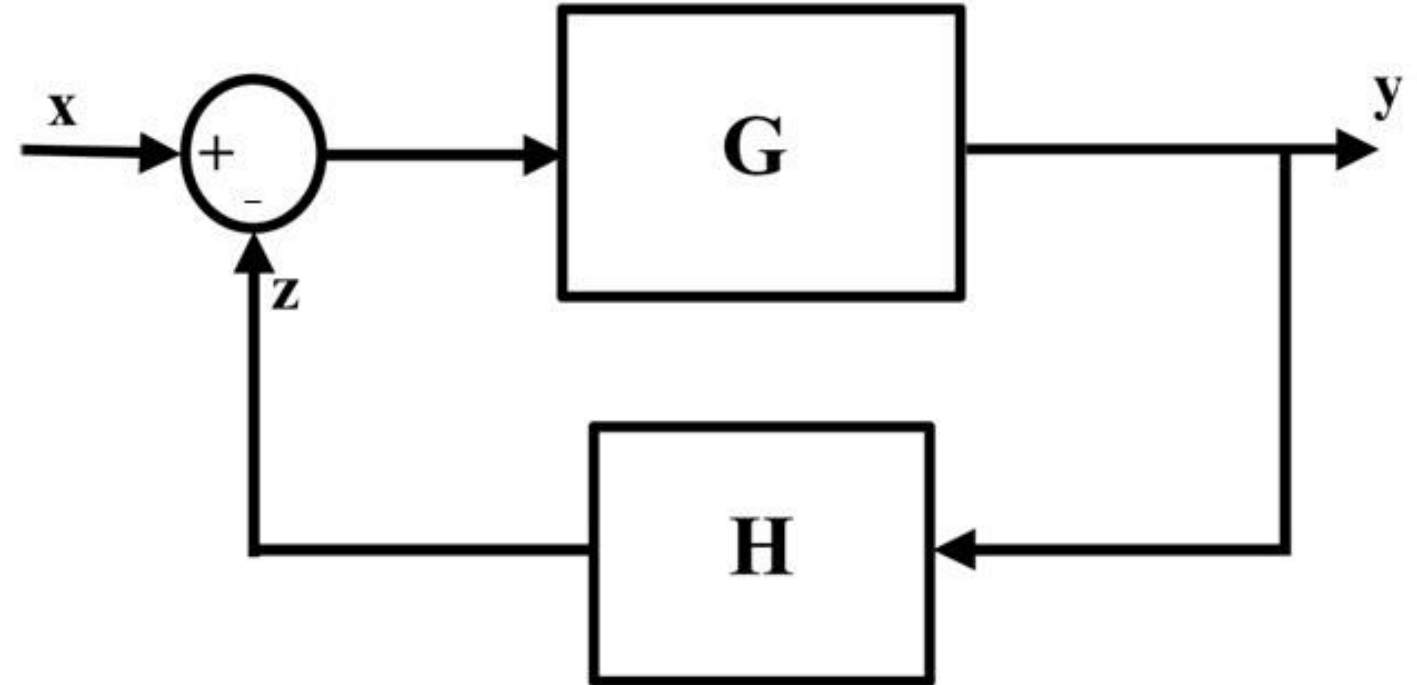




Control Systems I



HS 2026

About me:



Diogo Costa Pais



1st Semester Msc Mechanical Engineering



Languages: English, Italian



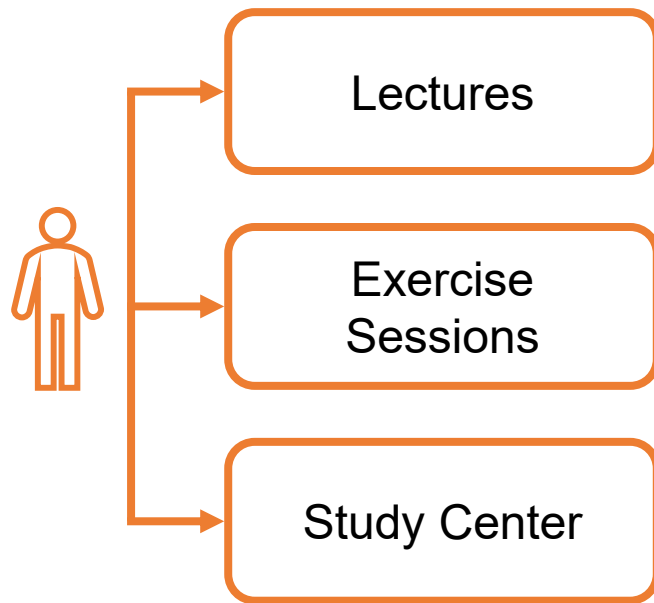
Contact: dcosta@ethz.com

All recitals are available here:

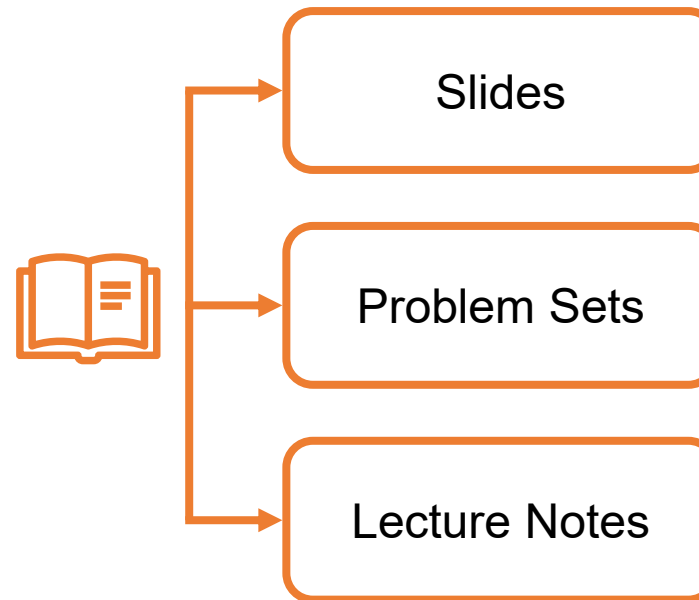


<https://dcostapais.github.io/dcosta.github.io/>

Course organization



Study Center Wednesdays ML D 28
(from 3rd week of the semester)





Recitation Structure

- 1) Quick recap of last week's recitation**
 - 2) Lecture recap**
 - 3) Past exam exercises**
 - 4) Homework time**
- Questions? Ask anytime!



Disclaimer for this course:

1. This course combines a lot of concepts from different lectures:
 - Linear Algebra II -> ODEs
 - Mechanics III -> Modeling
 - Analysis III -> Laplace Transformations
2. It may seem very abstract in the beginning
 - Solution: **Bare with me, it will eventually make sense**
3. I will always try to connect everything to examples to make the concepts easier to understand

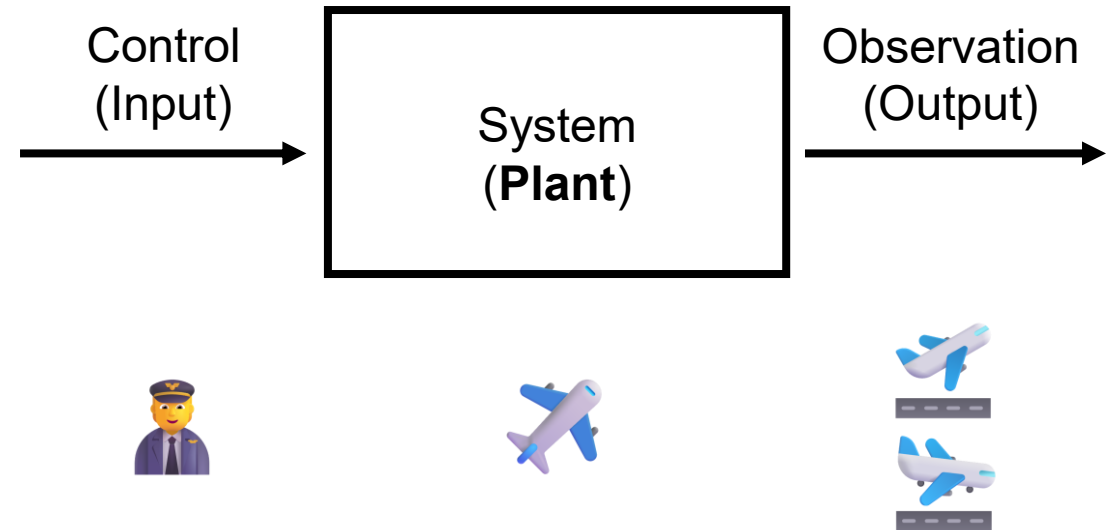
What is Control Systems?

We will start with a System (=Plant)

- A physical process/system we want to control
- Examples:
 - Car
 - Drone
 - Airplane
 - etc.

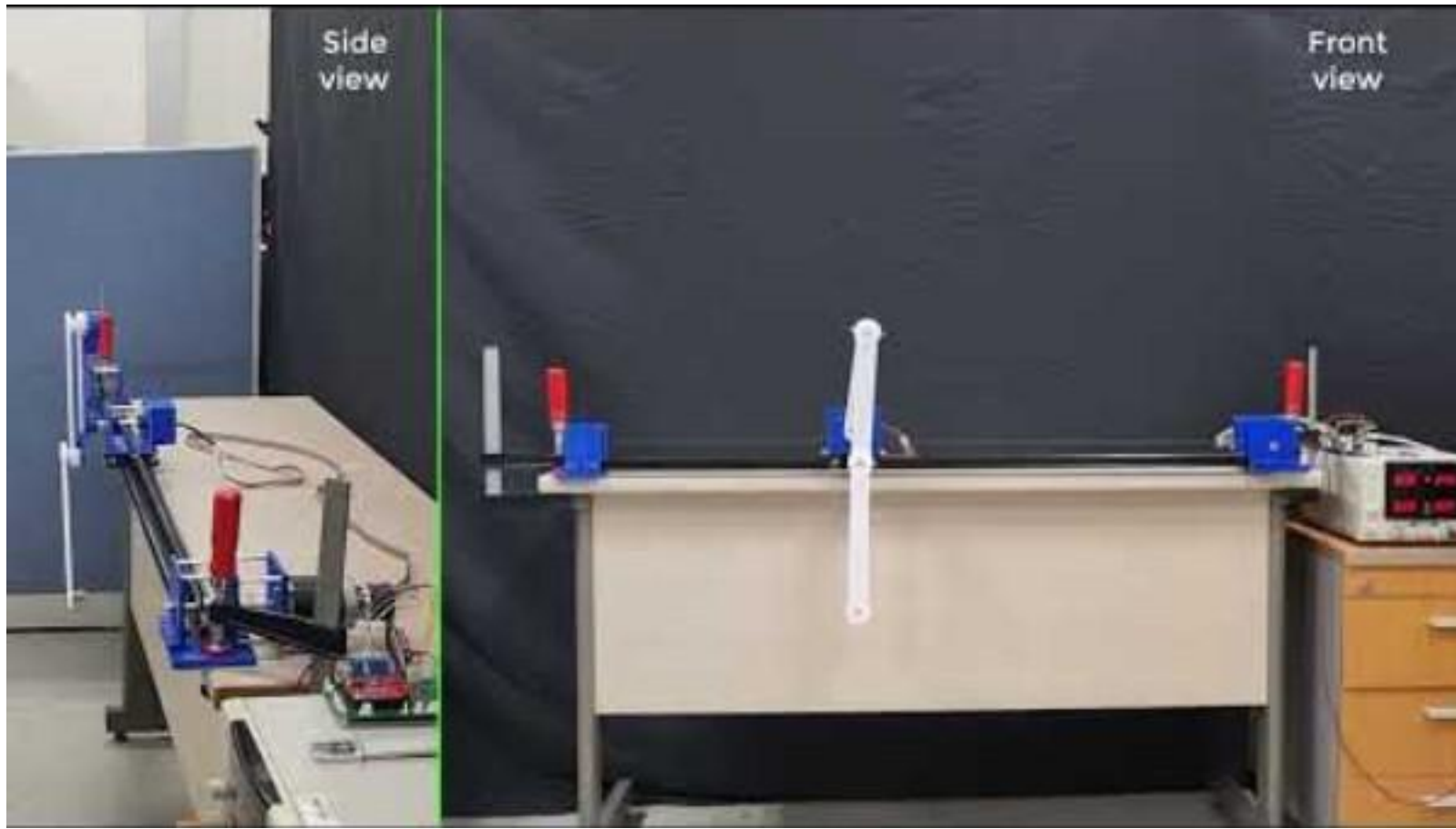
In this course:

-> Mostly linear SISO Systems
(single-input, single-output)



Goal: Given some reference the controller generates the needed input so that the output matches the reference

Example: inverted pendulum

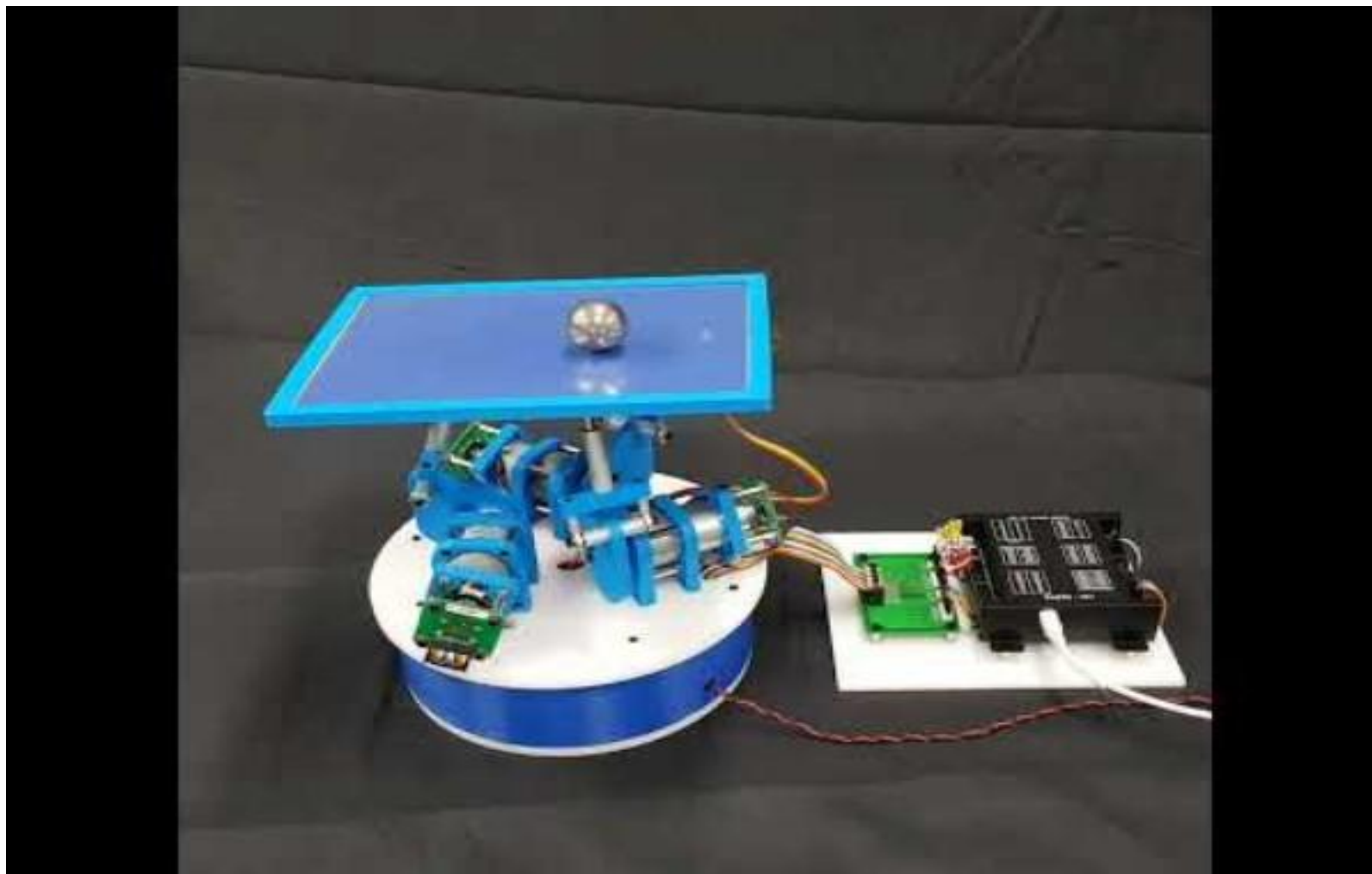


What are our control objectives?



- **Stabilization:** make sure the plant does not deviate too much from the desired behavior
- **Performance:** execute the desired task as accurately as possible
- **Robustness:** perform well even in the presence of disturbances, noise, etc.

More examples:





Course “chapters”:

- **Modeling:** learn how to represent a dynamic control system in such a way that it can be treated using mathematical tools
- **Analysis:** understand the basic characteristics of a system, and how the input affects the output. We will try to understand how a system will behave under feedback, given its open-loop behavior
- **Synthesis:** figure out how to modify a system, typically by feedback, in such a way that it behaves how we want

ONE LAST THING: EXAM

You will get a 1-sheet (2 A4 pages) summary sheet.

- Created last year through feedback with the students.
- This year: If you want to add something to the summary sheet, interact with the **weekly forum on moodle**

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ETH zürich
Course Catalogue

↓ Courses | ↓ Lecturers | ↓ Time and Place

Search →

151-0591-00L Control Systems I

Semester	Autumn Semester 2026
Lecturers	E. Frazzoli
Periodicity	yearly recurring course
Language of instruction	English

Up to 20 A4 sheets (40 A4 pages) of notes personally written by hand on paper